Introduction to Psycholinguistics

Unit 3

Spoken Word Recognition

(Contd.)

Lexical Access: Getting the Meaning

- The selection process discussed previously is really only about deciding which word-form we've heard, essentially, a decision that what we've heard is word-form X rather than word-form Y (e.g., 'captain' rather than 'captive').
- But word-forms (also called lexemes) aren't just bare sound-shapes sitting in isolation, they are linked to lemma-based content information: things like the word's meaning, its grammatical category (noun, verb, etc.), and other stored properties.
- **Lexical access is the term used specifically to refer to the point at which this lexically-stored content information actually becomes available to the listener.**

Parallel Contact vs. Serial Search: Two Competing Architectures

- Sometimes 'access' is used to mean the point at which lexical entries are **first contacted** (i.e., essentially equivalent to the 'contact' stage)
- 'access' refers specifically to the point where the content information (meaning, grammar, etc.) associated with a word becomes available.
- Many models simply assume that **as soon as** a word-form has been contacted, **all** of the information associated with that word becomes available essentially immediately, with no meaningful gap between 'contact' and 'access.'
- There's actually strong evidence that access can be demonstrated to happen at a specific, identifiable point, and that point turns out to be **earlier** than final word selection.

Access Happens Before Final Selection

- Lexical access clearly occurs before the selection of a single, unique candidate word.
- In other words, we get semantic/grammatical information about multiple candidate words before the recognition system has actually settled on which one of them is correct.
- Zwitserlood's (1989) priming experiment examined specifically for what it tells us about access:
  - participants heard the spoken prime captain (or the control prime justice), and were shown visual targets either early or late relative to the prime.
  - participants were able to respond more rapidly to both of the test targets- ship (related to captain) and money (related to capital, the never-spoken cohort partner), when they had heard only the onset fragment /kæp/ of the prime captain, compared to hearing an equivalent-length fragment of the unrelated control prime.

- The fact that just this brief onset fragment, /kæp/, is enough to produce facilitation for both semantically-related targets shows that this fragment is sufficient for the listener to make contact with the lexical representations for both captain and capital, and, critically, this contact must come along with access to the semantic content of both of these representations, since it is that semantic content (money-relatedness, ship-relatedness) that is doing the priming work on the visual targets.
- **The lexical access,** in the sense of retrieving a word's stored meaning and grammatical information**, is not something that waits until a single winning candidate has been selected out of the cohort**.
- Instead, **access happens in parallel**, for multiple still-competing candidates at once, mirroring exactly the same parallel, graded-activation picture established for contact and activation.

Recognition and Context Effects

- Cohort Model as described so far, lexical access is only made possible after a word has already been activated on the basis of the incoming speech signal, i.e., strictly on the basis of **bottom-up information**.
- **But** we often find that we're able to complete or continue someone else's sentence for them, before they've even finished saying it.
- This intuitively suggests that **we must be able to predict an upcoming word on the basis of top-down, contextual information**, i.e., using the meaning of the sentence so far to guess what's coming next.

Evidence AGAINST Simple Context Pre-Selection

'The men stood around for a while and watched their captain . . .'

Method: sentence contexts like above were used in cross-modal priming experiments, this time embedding the prime word within a full sentence context rather than presenting it as an isolated word.

Condition 1: target presented before the prime word begins: when a related visual target word (ship, related to the eventual prime word captain) was presented to participants immediately before they even started hearing the actual prime word captain within the sentence, there was NO facilitation for ship, relative to an unrelated control condition

- **Condition 2**: target presented partway through the prime: same target word was presented later, partway into hearing the actual prime word (specifically, after the onset fragment /kæp/ had already been heard), facilitation did appear.
- The conclusion:
- It appears that **bottom-up information (the actual phonetic input) has priority over contextual information** in the very earliest stages of making initial contact with stored lexical forms.
- Context alone, no matter how strongly constraining, does not seem able to pre-activate a specific candidate word ahead of any acoustic input actually arriving.

So Why Does It Feel Like We Can Predict/Finish Sentences? Reframing the 'Sentence-Completion' Intuition

- Just as individual words can prime other words that are related in meaning, in exactly the same way the interpretation of an entire sentence context can activate a broad 'field of meaning'- and this, in turn, can cause a whole set of related words to become more generally available or accessible.
- This contextual/semantic-field activation process is simply not as fast as genuine bottom-up acoustic activation, so it can create a real, useful sense of anticipation and predictability over the timescale of finishing a sentence, without actually being fast enough to pre-select a specific word candidate ahead of the incoming sound.

How Context DOES Genuinely Help, Just Not Via Pre-Selection

- Words are activated on the basis of bottom-up information
- Lexical access then makes available the lexical information (grammatical and semantic) associated with those activated candidate words
- This retrieved lexical information is then assessed against the listener's developing interpretation of the sentence so far.
- This assessment step results in changes to the activation levels of the various activated word candidates, context can boost the activation of candidates that fit and suppress the activation of candidates that don't, but only after those candidates have already been bottom-up activated in the first place

The key practical consequence of this integration process

- **The recognition point can move earlier than the uniqueness point**; because contextual and lexical information get integrated so rapidly, a word can actually be recognized earlier in a sentence context than it would be if the very same word were heard completely in isolation.

- Example: 'trespass'
- In isolation: for the word trespass, its recognition point is the same as its uniqueness point, this occurs at the /p/ sound.
- This is the specific point at which the remaining members of its word-initial cohort, words like tress and trestle, can finally be ruled out purely on acoustic grounds.
- In a constraining sentence context: *'The poacher was found guilty of tres*-...'In this context, the recognition point for trespass moves earlier, to the /s/ sound (i.e., before the /p/ is even reached).
- **Why**: the competing cohort members tress and trestle simply do not fit semantically/grammatically into this sentence context ('found guilty of tress...' and 'found guilty of trestle...' don't make sense), and so the activation levels of these competitors' lexical representations will have already dropped away, due to the contextual-integration process described above, well before the acoustic signal alone would have ruled them out.

Not All Contexts Constrain Equally

- The Graded Constraint Experiment (Marslen-Wilson, Brown & Tyler, 1988)
- **Method**: this is a word-monitoring experiment, using spoken sentences. The same target word- guitar, was embedded as the object of different sentences, and participants across different groups were simply told to listen out for the target word and press a response button the instant they heard it.
- Response times (RTs) were measured starting from the onset of the word 'guitar' itself.

The four sentence contexts used, each varying only in the preceding verb, presented in order of decreasing plausibility:

- **(8.3) 'The crowd was waiting eagerly. The young man grabbed the guitar . . .'**

a guitar is something that is very plausibly grabbed. Mean RT = 241 msec (averaged across a set of similar example sentences).

- **(8.4) 'The crowd was waiting eagerly. The young man buried the guitar . . .'**

a guitar can be buried, technically, but this is a somewhat unusual thing to do. Mean RT = 268 msec.

- **(8.5) 'The crowd was waiting eagerly. The young man drank the guitar . . .'**

The verb drink can take an object noun phrase, but that object noun phrase normally has to denote a liquid, and guitars, of course, are not liquid. Mean RT = 291 msec.

- **(8.6) 'The crowd was waiting eagerly. The young man slept the guitar . . .'**

the verb sleep is intransitive, it simply does not take an object at all, grammatically. Mean RT = 320 msec (the slowest of the four)

- Response times got progressively and increasingly longer as you move from sentence (8.3) through to (8.6), this directly reflects the steadily changing contextual constraints across these four sentences, constraints that are determined entirely by the choice of verb (grabbed → buried → drank → slept).
- The more implausible or ungrammatical the fit between the verb and 'guitar' as its object, the slower participants were to recognize/detect the target word, even though the acoustic signal for the word 'guitar' itself was, of course, identical across all four conditions.

- Word-monitoring times were also found to generally **decrease** the **later** a target word appeared within a sentence, **as long as** that sentence remained grammatically well-formed.
- **Interpretation:** this shows that as we hear progressively more words within a sentence, we're able to build up an increasingly **complete interpretation** of the surrounding context, and that richer, more complete context becomes progressively **more constraining** on which words could plausibly occur next, reinforcing the point that context operates through a continuous, integrative process rather than through simple, one-off pre-selection.

Neuroimaging Corroboration (Kuperberg et al., 2000)

- **Method**: this study also used sentence sets structured just like (8.3)–(8.6) above, i.e., varying the plausibility of the verb-object fit.
- This time, however, participants simply had to listen to each entire sentence and then judge whether or not it 'made sense.

- **Finding 1**: a shared/common activation pattern: all three of the violation conditions (corresponding to sentence types 8.4, 8.5, and 8.6, the increasingly implausible/ungrammatical ones) showed increased activation in a particular brain area. Critically, this same brain area has independently been associated, in studies of brain-damaged patients, with comprehension difficulties, suggesting this region plays a general role in language **comprehension breakdown/difficulty**.

- (8.3) 'The crowd was waiting eagerly. The young man grabbed the guitar . . .'
- (8.4) 'The crowd was waiting eagerly. The young man buried the guitar . . .'
- (8.5) 'The crowd was waiting eagerly. The young man drank the guitar . . .'
- (8.6) 'The crowd was waiting eagerly. The young man slept the guitar . . .'

- **Finding 2**: distinct, violation-type-specific activation: in addition to that shared area, separate, distinct areas of increased brain activity were also found that were specific to each of the different types of violation (i.e., the 'unusual-but-possible' violation in 8.4 activated somewhat different areas than the 'wrong argument type' violation in 8.5, which in turn differed from the 'no object allowed at all' violation in 8.6)

- (8.3) 'The crowd was waiting eagerly. The young man grabbed the guitar . . .'
- (8.4) 'The crowd was waiting eagerly. The young man buried the guitar . . .'
- (8.5) 'The crowd was waiting eagerly. The young man drank the guitar . . .'
- (8.6) 'The crowd was waiting eagerly. The young man slept the guitar . . .'

- It is conjectured that the **shared** area, the one showing increased activation across all three violation types, is specifically the brain region responsible for the general process of **integrating information**, which is, of course, exactly the kind of integrative process as the mechanism by which sentence context influences word recognition.

Frequency, Competition, and Neighbourhoods

- There are consistent frequency effects in word recognition (Forster & Chambers, 1973; Monsell, 1991; Murray & Forster, 2004), that is, **words we encounter more often in everyday language have a genuine processing advantage** over words we encounter rarely.
- This advantage shows up empirically as faster response times and greater accuracy for common words in tasks such as lexical decision (where participants judge whether a sound is a real word or not).

Frequency Effects: The Core Phenomenon

- this frequency effect can be illustrated by modifying the earlier activation-over-time picture captured now as a slower rate of increase in activation for the lower-frequency words captive and capital, relative to the higher-frequency word captain.
- The result: even at a point where the phonetic input still matches all three words equally well (e.g., at the shared fragment /kæpt/), captain already shows an activation advantage over its rivals, purely because of its higher frequency, not because of any difference in the acoustic input itself.

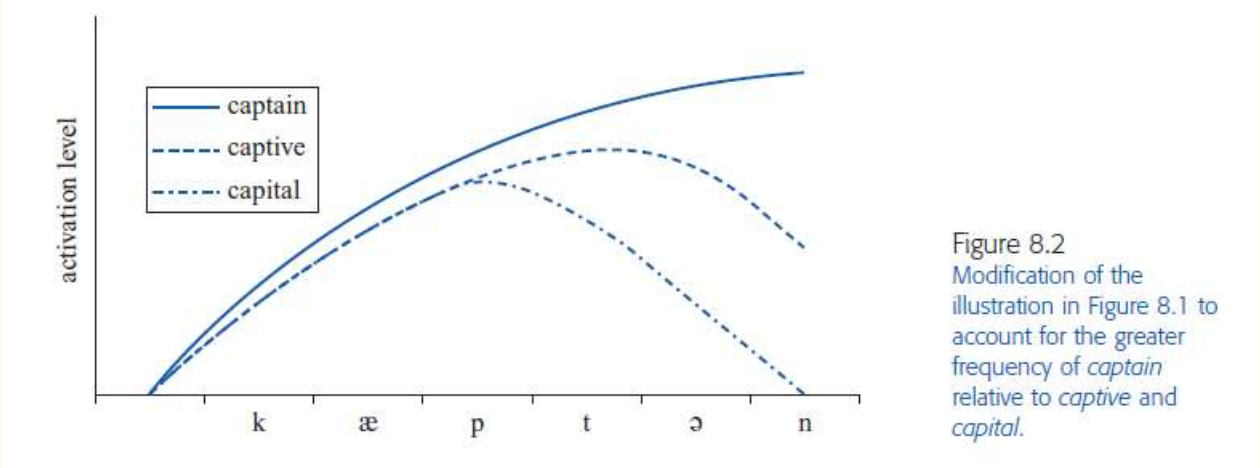


Figure 8.2
Modification of the illustration in Figure 8.1 to account for the greater frequency of *captain* relative to *captive* and *capital*.

Two Competing Mechanisms for How Frequency Effects Arise

- **Mechanism 1**: differential rate of activation increase in Figure 8.2 suggests frequency effects could simply be a matter of how quickly activation rises for different candidates; frequent words' activation rises faster per unit of matching input than rare words' activation does.
- **Mechanism 2**: different 'resting levels' of activation: an alternative proposed mechanism is that **different words have different baseline/resting levels of activation even before any input arrives at all**. Frequent words simply start with a higher resting level than rare words, giving them a built-in head start.
- How this mechanism explains the effect: if we assume the same input produces the same rate-of-change in activation for all word candidates (i.e., no difference in the speed of the rise), then a resting-level advantage for frequent words will simply persist throughout processing

Recency Effects: A Closely Related Phenomenon

- Recency is described as words that we have encountered quite recently seem to gain, at least temporarily, a processing advantage over similar words that haven't been heard for a while (Oliphant, 1983).
- Proposed mechanism: if actually using/encountering a word causes a temporary increase in its activation level, and if it then takes some amount of time for that activation to decay back down to its normal resting level, then encountering that same word a second time, while its activation is still elevated above baseline from the first encounter, would naturally result in faster recognition the second time around, essentially 'catching' the word while it's still artificially activated from before.

Neurophysiological (ERP) Confirmation

- In neurophysiological studies, frequency effects have been shown to be reflected in ERP (event-related potential) measures, specifically, in the size of the electrical brain response to words, as well as in the speed of certain components of that electrical response (see Kutas & Federmeier, 2007, for a review).
- The interpretive takeaway: both of these ERP findings (larger and faster responses for higher-frequency words) point in the same direction, they indicate greater neural stimulation occurring in response to higher-frequency words, providing a physiological/biological correlate to the purely behavioral (reaction-time-based) frequency effect described before.

Competition Between Words

- Recognizing a word depends not only on the properties of that word itself (its sound-form, its own frequency, and so on), it **also depends on the properties of the other words that compete with it**.
- We already know that nasalization information on a vowel in a word like soon helps listeners distinguish it early from similar-sounding words like suit or soup. Given the input /sũ.../, listeners may already be forming a strong hypothesis that the word is specifically 'soon,' and not 'suit' or 'soup,' well before the final consonant arrives.
- Informativeness depends on what competitors actually exist: the very same nasalization cue has a completely different informational value if it instead occurs on the vowel in the word fruit, because there simply is no English word that has a nasal consonant following the /u/ vowel in that particular phonetic environment. So if nasalization shows up on the vowel in 'fruit,' it doesn't tell the listener anything useful about word identity at all.

- The informativeness of any given property of the speech signal depends entirely on what alternative possibilities exist, and on whether those alternatives can actually be ruled in or out by that particular property.
- This is referred to as the notion of 'contingency of choice', for example, knowing that you've heard the word cat depends not merely on the sounds that make up 'cat' in isolation, but also, implicitly, on knowing that you have not heard the competing words cap, can, cash, and so on (Warren & Marslen-Wilson, 1987, 1988).
- **Recognition is inherently a relative, competitive judgment, not just a positive match to one stored form**.

- **Why this all matters? The need to model relationships between words, not just words in isolation:** these kinds of considerations make clear that a proper model of word recognition needs to take into account not just individual, isolated words, but also the relationships those words have with their competitors. This notion of competition has become a central, extensively studied concept in models of word recognition.
- **Modeling competition as inhibition:** one of the most widely-adopted ideas is that the practical result of competition between two words is that the words inhibit one another, that is, competition has a genuinely negative impact on the activation levels of rival/competitor words (not just a passive failure to help them, but an active suppressive effect).

- **The proposed size-dependency of this inhibition:** the actual size of this inhibitory effect might depend on the relative activation levels of the words that are currently in competition with one another.
- **Example:** captain vs. captive again: given an ambiguous input like /kæpt/ (which still matches both words), the more frequent word captain will already have a higher activation level than the less frequent word captive (per the frequency-effect discussion earlier), and this greater activation level allows captain to exert a stronger inhibitory influence on captive than captive is able to exert back on captain.
- **The net effect**: this inhibitory dynamic effectively amplifies/boosts the activation gap that already exists between the two words, making it even more likely that captain will ultimately be the one recognized, especially in the absence of any further differentiating acoustic information arriving later. Competition thus acts as an amplifier on top of the base frequency advantage, not just a neutral backdrop.

Neighborhoods: A Broader Notion of Competition

- What is a 'neighborhood'? A word belongs to a neighborhood of words that share similar phonological properties with it (Luce & Pisoni, 1998; Vitevitch, Stamer & Sereno, 2008).
- A word-initial cohort is itself technically one type of neighborhood- but researchers have also studied more extensive neighborhoods that go beyond simple shared-onset relationships.
- **Neighborhoods consisting of all words that differ from the target word by just one phoneme**. Critically, this '**one phoneme difference**' can occur anywhere within the word (beginning, middle, or end), regardless of position, unlike a cohort, which is specifically restricted to shared word-onsets only.
- the full neighborhood of 'money,' as given in the text: the following words all differ from money by exactly one phoneme: many, meanie, moony, bunny, funny, gunny, honey, runny, sonny, sunny, tunny.

- **What neighborhood properties actually affect, and how?**
- The recognition of a target word, measured in tasks such as lexical decision and naming (i.e., reading a word aloud), is affected by three interacting properties of its neighborhood:
  - How many neighbors the word has (the sheer size/count of its neighborhood).
  - The degree of phonetic similarity of these neighbor words to the target word (i.e., how much overlap of form there actually is).
  - The frequencies of the words that make up that neighborhood.

The key empirical generalization:

- A word that has a lot of high-frequency neighbors is identified more slowly and less accurately than a word that has only a few neighbors, or that has only low-frequency neighbors.
- In other words, **living in a 'crowded,' competitive phonological neighborhood**, especially one crowded with common, high-frequency competitor words, **actively hurts recognition speed and accuracy for the target word**, because all those neighbors are simultaneously being activated and competing for selection.

- Terminology: 'neighborhood density': these combined factors (number, similarity, and frequency of neighbors) together constitute what is commonly referred to as the word's *neighborhood density*.
- Demonstrated real-world impact: neighborhood density has been shown to have a measurable influence on both:
  - Word recognition (Vitevitch & Luce, 1998), and
  - Word production (Vitevitch, 2002)
- showing that this isn't purely a comprehension-side phenomenon, but affects the broader lexical system, including how we retrieve words to speak them ourselves.

Recognizing Morphologically Complex Forms

Two Types of Morphology

- **Inflectional morphology**: this involves **adding affixes purely to mark grammatical information**, it does not create a new kind of word, just a grammatically-marked variant of the same underlying word. Examples
  - Plural marking: cat → cats
  - Past tense marking: jump → jumped
  - Subject-verb agreement marking: he swims vs. I swim (no -s)
- **Derivational morphology**: this involves **adding affixes to actually create a different kind of word** altogether, often changing its grammatical category. Example:
  - Kind → kindness
  - Dispose → disposition

Productivity differences, and the resulting irregularities:

- Inflections tend to be more productive than derivations overall, though there is real variation within each of these two broader categories too, not just between them.
- making an English plural by adding -s, as in cats, is an extremely common, highly productive pattern, with plentiful examples. By contrast, making a plural by adding -ren and simultaneously changing the internal vowel, as in child → children, is a genuinely unusual/irregular pattern, there just aren't many other words that form their plural this way.

Processing Inflections

The proposed rule-governed recognition process, mirroring production:

- just as it's claimed that the language production system builds the form cats by combining the base word cat with the highly productive affix -s, following a general rule like 'to make a plural, add -s to the stem', so too, symmetrically, the word recognition system would work in reverse: it would disassemble the incoming input cats through a process called **morphological decomposition,** stripping the recognized form back down into its component base + affix.

Why this fits naturally with the left-to-right processing model already established:

- This decomposition process is described as reasonably straightforward, specifically because of the 'left-to-right' processing system assumed throughout this chapter for spoken language.
- Since the base word cat would, in any case, already get contacted purely on the basis of hearing its first three phonemes (which are identical whether the word turns out to be 'cat' or 'cats'), the base-word contact happens 'for free,' essentially, before the affix is even reached.

A cross-linguistic generalization supporting this left-to-right-friendly design:

- Broader surveys of the morphological characteristics of the world's languages suggest that accessing an inflected form via its stem may actually be a very common cross-linguistic strategy.
- This is because inflectional affixation, across the world's languages, is predominantly done through suffixes (i.e., adding material onto the end of a word), rather than through prefixes (added to the beginning) or infixes (inserted somewhere in the middle).
- Since suffixes come after the stem, a listener processing left to right naturally reaches the (unmodified) stem first, which aligns perfectly with easy stem-based access.

The Combined-Frequency Finding

- There is considerable evidence, from both spoken and visual word recognition studies, concerning exactly how we process inflected word-forms.
- A crucial refinement to the simple 'frequency effect': it turns out that the frequency statistic that actually matters for predicting recognition speed is not simply the frequency of the bare base form alone (e.g., just counting occurrences of cat by itself), instead, it is the combined frequency of the base form together with all of its inflected forms (i.e., cat and cats counted together as one pooled total) that turns out to be the better predictor of recognition time.
- This pattern suggests that every time we encounter an inflected form of a word (like cats), we are, in some functional sense, also simultaneously encountering the base word itself (cat), with that encounter effectively adding to the base word's own frequency count in the mental lexicon's bookkeeping.

Inflected Forms Are Harder to Access in Isolation

- It has been shown that inflected forms are actually harder to access than one might expect, specifically when presented out of context.
- Example: the isolated spoken fragments /deɪz/ and /pækt/ tend to get reported/perceived by listeners as the uninflected words daze and pact, respectively, rather than as the inflected forms days and packed, even though, notably, the inflected forms (days, packed) are actually higher in frequency than the uninflected competitors (daze, pact).

Regular vs. Irregular Inflection

- There is clearly a real processing difference between regular and irregular inflection, and multiple lines of evidence converge on this:
- Developmental evidence: children tend to regularize irregular forms (producing foots instead of feet), which itself suggests that irregular forms, even ones that are actually quite frequently encountered (like went, the irregular past tense of go), simply do not have as strong a hold on the developing child's processing system as the productive regular pattern does.

Repetition priming evidence in adults:

- **Baseline comparison:** equally strong priming is found for identical prime-target pairs, that is, the word pour appearing again after being preceded, several items earlier, by that exact same word pour - compared against a baseline condition without any such earlier repetition.
- **The regular-inflection test case:** that same strength of priming is also found in a condition where the target pour is instead preceded not by the identical word pour itself, but by its regularly inflected form, poured. In other words, poured primes pour just as strongly as pour primes pour, showing an extremely tight link between a regular inflected form and its base.
- **Contrast with the irregular case:** compare this against the finding that, although there is some priming of the base swim by its irregular past-tense form swam, this priming effect is not as strong as the priming of swim by swim itself (i.e., not as strong as the identical-repetition condition). This shows that the irregular form swam is functionally less tightly bound to its base swim, in processing terms, than a regular form like poured is to pour.

ERP (brain-imaging) confirmation of this dissociation:

- These processing differences between regular and irregular forms are also directly reflected in ERP studies.
- These have shown that different brain areas are activated for the two types (Newman, Ullman, Pancheva, Waligura & Neville, 2007).
    - Regular forms show activation specifically in brain areas known to be involved in general rule processing
    - irregular forms instead lead to activation of areas that are associated with lexical storage (i.e., areas linked to retrieving whole, memorized items rather than applying a rule)

Processing Derivations

- Combined-frequency effects extend to derivations too, but selectively: turning now to derived words, it has been found that recognition of a word like account is affected by the combined frequency of words that are related to it via some shared base meaning, for example, accountant and unaccountable,
- but, importantly, recognition of account is NOT similarly affected by other words that are merely superficially/formally similar but are not actually semantically related in this derivational-family sense, the text's specific examples being count, counter, and viscount (all of which share letters/sounds with 'account' but aren't meaningfully derived from the same root in the relevant sense).

Derivational priming exists, but is weaker than inflectional priming:

- It has also been found that derived words like ability can prime their base forms, like able, but, notably, this priming effect is not as strong as the priming effect found between inflected forms and their bases (e.g., not as strong as the poured→pour effect described earlier).
- This establishes a rough hierarchy of priming strength:

  identical repetition > regular inflection > derivation.

Transparent vs. Opaque Derivational Relationships

- priming effects between a base form and a morphologically related derived word exist only if the meaning relationship between the two words is transparent, i.e., clear to ordinary language users.
- example pair that DOES prime, transparent relationship: mature and maturity do prime each other.
- example pair that does NOT prime, opaque relationship: casual and casualty do NOT prime each other. (Marslen-Wilson, Tyler, Waksler & Older, 1994.)
- Why this particular result makes intuitive sense

since we would not generally imagine that most ordinary language users are consciously aware of, or actively represent, the historical/semantic relationship between 'casual' and 'casualty' (their modern meanings have drifted apart quite a bit, even though they're etymologically related), this lack-of-priming result is not surprising, the derivational relationship, while real historically, is simply not 'live' or transparent for most speakers today.

The Prefix/Suffix Asymmetry

- Prefixed words DO prime other, related prefixed words, in both directions:
- example 1: disloyal primes loyal, and vice versa (loyal also primes disloyal), priming works both ways between a prefixed form and its base.
- example 2: prefixed words also prime other prefixed words that are morphologically related to them (not just their bare, unprefixed base), specifically, remount primes dismount, even though neither is the 'bare base' of the other; they are both prefixed relatives of the same root (mount).
- BUT suffixed words do NOT prime other, morphologically-related suffixed words: the specific example given is that maturity does NOT prime maturation, even though both are legitimate suffixed derivations sharing the same root/meaning family.

The explanation offered for this asymmetry

- any priming effect that might otherwise be expected to arise from the genuine morphological relatedness of maturity and maturation is proposed to be offset/cancelled out by competition that arises between them.
- Specifically because maturity and maturation are both members of the same word-initial cohort (they share the same onset sounds, 'matur-,') and so are direct competitors for contact/activation in exactly the sense established back)
- The competitive inhibition between same-cohort rivals effectively cancels out whatever positive morphological-relatedness priming there might otherwise have been.

Why the mature/maturity pair is then presented as a genuinely special case:

- since priming is, in fact, still found for the mature/maturity pair, despite the fact that mature and maturity also, in principle, share cohort-competitor status with each other in a similar way, it would seem that the specific relationship between a base form and a word derived directly from that exact base is somehow 'rather special', strong enough to survive/overcome the same kind of cohort-competition inhibition that cancels out the maturity/maturation effect. This is presented as a genuinely interesting, unresolved exception in the data.

Words and Rules: The Big Theoretical Debate

The two competing overall architectures for the mental lexicon, as applied to complex-word recognition.

- A rule-based recognition system: under this view, an input word gets analyzed into its meaningful constituent parts on the fly (e.g., recognizing cats by decomposing it into cat + -s, applying a general rule).
- A full-listing system: under this alternative view, every word, whether it is morphologically simple or complex, is simply listed as its own separate entry in the mental lexicon, with no online decomposition process needed at all (Marslen-Wilson, 2007).

Overall Conclusion

- Taking together all of the evidence surveyed across this whole section, the processing differences found between regular and irregular inflected forms, the differences found between transparent and opaque derived forms, and the genuinely complex pattern of interactions found across various priming studies (including the prefix/suffix asymmetry above), the overall suggestion is that the **mental lexicon probably relies on BOTH systems simultaneously, rather than exclusively on just one or the other.**
- This dual-mechanism conclusion is further supported by the neurophysiological (ERP) studies, specifically, the finding that regular forms activate rule-processing brain areas while irregular forms activate lexical-storage brain areas is essentially direct physiological evidence for exactly this kind of dual-route architecture.

- Rule-based analysis of productive and regular affixation is understood to exist alongside stored representations of complex forms that do not lend themselves to easy morphological analysis (i.e., irregular and/or semantically opaque forms).
- The mental lexicon, in other words, is not a single, uniform system, it's a hybrid architecture that applies rules where the pattern is productive and transparent, and simply stores whole forms where it isn't.